

CHOCKALINGAM BALAN

Senior Software Engineer — Backend, Data, Fullstack & AI

Chennai, India | (+91)-9524031307 | chockalingam1805@gmail.com

<https://chocka-devfolio.vercel.app/> | <https://www.linkedin.com/in/chockalingam-balan/>

SUMMARY

Senior Software Engineer with 5+ years of experience architecting scalable backend platforms, distributed ETL execution engines, and fullstack ML systems deployed across global markets. Proven expertise designing DAG-based orchestration engines, multi-tenant backend architectures, and ML deployment automation platforms using FastAPI, Django, Databricks, and OpenAI. Strong experience building high-throughput APIs, distributed data pipelines, and production-grade AI-powered systems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, Django, Flask, REST APIs, Microservices, Async Processing

Data Engineering: Databricks, PySpark, Pandas, SQLAlchemy, ETL Pipelines

ML / AI: OpenAI API, Claude Sonnet, Claude Opus, AWS Bedrock, Databricks ML, LLM Integration, AI-Powered Code Generation, AI/LLM Pipelines, XGBoost, LightGBM, H2O.ai

Frontend: React.js, Angular

Databases: SQL Server, PostgreSQL, MySQL, Databricks Hive

Cloud / Infrastructure: Azure, Docker, AKS, GCP, AWS Databricks

Orchestration: Apache Airflow, Celery

PROFESSIONAL EXPERIENCE

iLink Digital Inc — Technical Specialist | Chennai, India | Dec 2025 – Present

Project: Enterprise BI Migration Platform

- Architected a DAG-based ETL execution engine converting Tableau Prep flow JSON into an intermediate directed acyclic graph (DAG) representation, enabling deterministic and scalable execution of transformation workflows.
- Developed AI/LLM-powered migration accelerators to analyze Tableau Prep flow JSON and Tableau Data Source (TDS) XML and generate equivalent Databricks notebooks and M/Query transformations.
- Integrated LLM capabilities using OpenAI, Claude, and AWS Bedrock to interpret complex Tableau transformation logic, data-source metadata, dependencies, and business rules for automated target-platform code generation.
- Built reusable AI-powered pipelines for automated generation of Databricks notebooks and M/Query from Tableau workflow definitions, accelerating the modernization of existing Tableau data workflows.
- Designed an AI-powered automated unit-testing and validation pipeline to compare migrated Tableau logic with generated Databricks and M/Query outputs and automatically identify transformation discrepancies.
- Designed and implemented transformation parsing and compilation layer converting Tableau operations including joins, unions, aggregations, calculated fields, filters, and nested expressions into executable Spark and Pandas transformations.
- Built execution planning engine implementing dependency resolution and topological sorting algorithms to ensure correct ordering and execution of transformation nodes based on lineage and dependencies.
- Developed FastAPI orchestration service responsible for workflow ingestion, DAG generation, execution planning, execution triggering, job monitoring, and lifecycle management.
- Implemented asynchronous execution architecture using background task orchestration enabling concurrent execution of multiple workflows and improving throughput and scalability.
- Optimized orchestration and planning pipeline achieving API response time under 5 seconds for DAG generation and execution planning.
- Engineered OpenAI-based fallback execution engine enabling dynamic execution of unsupported or complex Tableau transformations that could not be parsed or executed deterministically using Spark or Pandas.
- Enabled dynamic interpretation and execution of nested and complex transformation expressions using OpenAI, significantly improving platform transformation coverage.
- Integrated Azure Databricks as the distributed execution engine enabling scalable processing of large-scale enterprise data pipelines.

- Integrated Azure Blob Storage for scalable storage of input data, output artifacts, workflow definitions, and execution results.
- Designed SQL Server schema for storing workflow metadata, execution status, transformation definitions, and execution tracking.
- Designed modular backend architecture following clean architecture principles ensuring maintainability, extensibility, and separation of concerns.
- Enabled automated migration of complex Tableau workflows to scalable distributed ETL pipelines.

Project: AI Claims Adjudication Platform

- Designed and developed FastAPI-based backend service to ingest insurance claim documents and orchestrate AI-based claim adjudication workflows.
- Built configurable backend architecture supporting structured request and response schema enabling scalable integration with upstream and downstream systems.
- Implemented asynchronous execution model enabling concurrent claim processing and improving throughput.
- Refactored backend processing pipeline reducing claim adjudication response time from 25 seconds to 10 seconds.
- Designed AI inference workflow enabling structured extraction, evaluation, and classification of insurance claim data.
- Containerized backend using Docker enabling consistent deployment and environment portability.
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Project: Nucor — Enterprise React Platform Refactoring

- Redesigned routing architecture by separating route definitions into dedicated routing module improving scalability and maintainability. Implemented protected route architecture ensuring secure access control across application components.
- Implemented lazy loading and React Suspense improving frontend performance, reducing initial load time, and enhancing user experience.
- Modularized session storage and application state handling improving maintainability and application architecture.

Tiger Analytics — Machine Learning Engineer | Chennai, India | Jan 2025 – Nov 2025

Project: Peplris — Retail Computer Vision Mobile Platform

- Designed and developed Django-based multi-tenant backend platform deployed across 10–12 international markets using a single shared codebase.
- Architected tenant-isolated database schema ensuring deployment, configuration, and schema updates for one market did not impact other markets.
- Built scalable REST APIs supporting mobile application workflows, analytics retrieval, and reporting.
- Automated ML model deployment pipeline enabling seamless promotion of trained models across environments.
- Implemented dynamic model credential and version management system enabling runtime model switching without redeployment.
- Reduced QA validation cycle time by over 90 percent by enabling direct model validation workflows.
- Implemented Celery-based asynchronous task execution framework enabling scalable execution of ML inference and background processing.
- Designed and implemented distributed ETL pipelines using Azure Databricks and PySpark enabling scalable data processing.
- Migrated legacy Pandas-based pipelines to distributed Spark pipelines improving scalability and performance.
- Built configuration-driven pipelines enabling reusable deployment across multiple international markets.
- Led go-live deployment and production rollout for multiple international markets.
- Supported analytics workflows across 10–12 international markets using a single scalable backend platform.

Tiger Analytics — Senior Analyst | Chennai, India | Oct 2023 – Dec 2024

Project: Peplris — Retail Computer Vision Mobile Platform

- Developed Django REST APIs supporting analytics workflows, KPI retrieval, reporting, and data ingestion.
- Implemented API versioning enabling backward compatibility and safe rollout of new features.
- Designed SQL Server database schema supporting scalable multi-market architecture.
- Built ETL pipelines using Pandas and SQLAlchemy supporting scalable data ingestion and transformation.
- Implemented database connection pooling improving performance and resource utilization.
- Optimized database queries improving backend performance and API response time.

KAAR Technologies — Fullstack & AI Application Developer | Chennai, India | Nov 2020 – Oct 2023

Project: Model Builder — ML Training Platform

- Developed React frontend enabling users to configure, train, monitor, and manage ML models.
- Developed Flask backend supporting ML model training, prediction, and lifecycle management.
- Designed backend services supporting scalable ML workflow execution.
- Integrated H2O.ai AutoML enabling automated model training and baseline forecasting.
- Engineered custom AutoML pipelines using XGBoost, LightGBM, and Gradient Boosting reducing dependency on proprietary AutoML tools.
- Designed and implemented Apache Airflow DAGs orchestrating model training, prediction, and evaluation workflows.
- Built automated model retraining and execution pipelines.
- Analyzed model performance using Power BI and delivered optimized models to business teams.
- Enabled automated ML training, forecasting, and evaluation workflows for business decision-making.

Project: Scenario Builder — Forecast Simulation Platform

- Developed React frontend enabling configurable scenario-based forecasting simulation.
- Developed Flask backend supporting scenario execution, prediction generation, and workflow orchestration.
- Enabled execution of ML forecasting pipelines supporting scenario simulation.
- Integrated Databricks Hive, GCP Bucket, and Firestore enabling scalable data storage and processing.
- Built Airflow DAGs enabling automated and manual execution workflows.
- Implemented asynchronous execution improving scalability and performance.

EDUCATION

Bachelor of Engineering — Computer Science

KLN College of Engineering, Sivagangai | Graduated: 2020